



MACHINE LEARNING–BASED DIAGNOSIS OF DIABETES

MINOR PROJECT PRESENTATION

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF
MASTER OF COMPUTER APPLICATIONS (MCA)

SUBMITTED BY:

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UNDER THE GUIDANCE OF:

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PROJECT RESPONSIBILITIES / TASKS

- Analysis of the problem related to early diagnosis of diabetes.
- Collection and study of the Pima Indians Diabetes Dataset.
- Data preprocessing including handling missing and invalid values.
- Implementation of machine learning algorithms using Python.
- Development of an interactive web application using Streamlit.

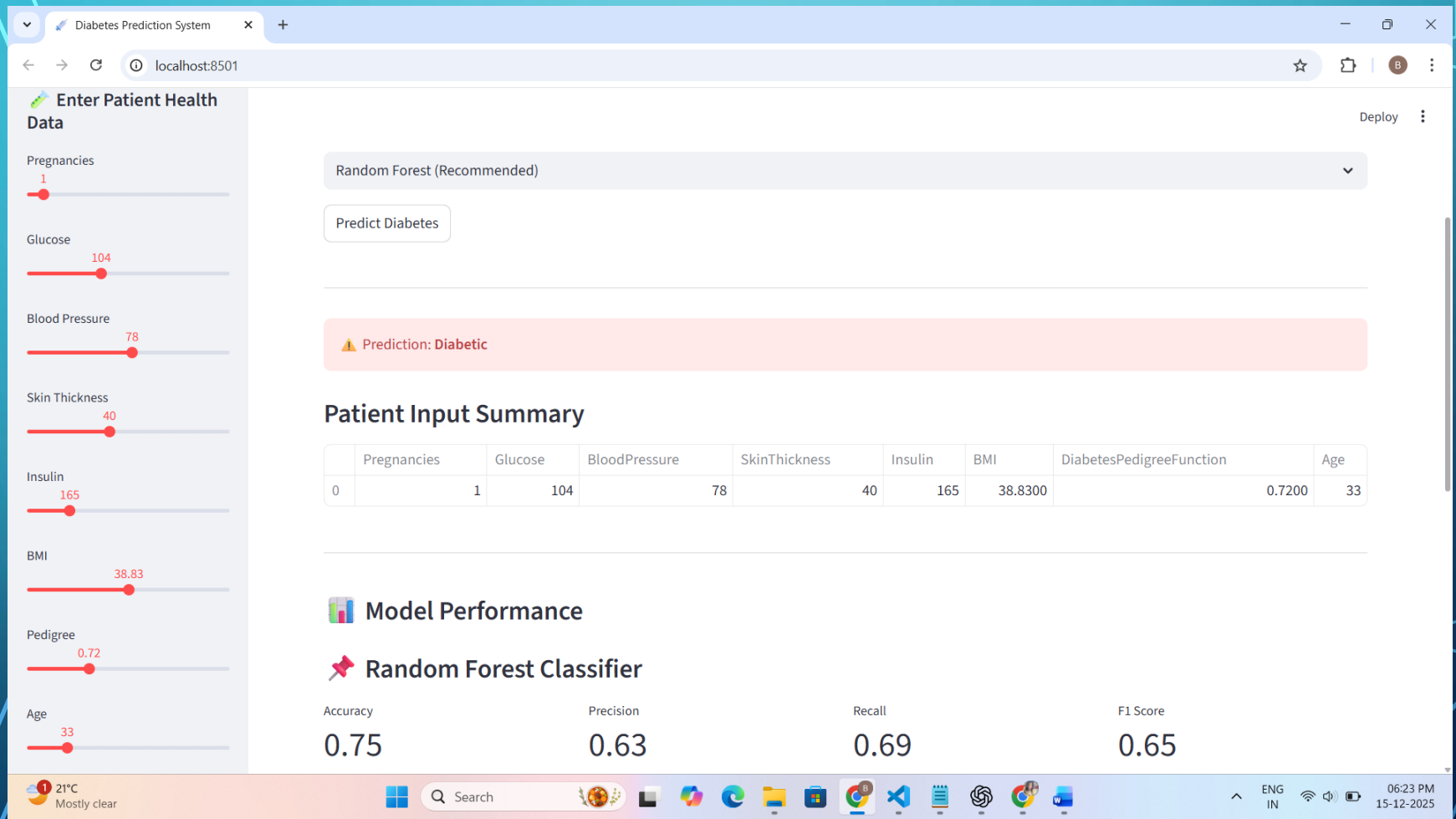
WHAT WE LEARNED

- Practical application of machine learning techniques in healthcare.
- Importance of data preprocessing in medical datasets.
- Model training, testing, and evaluation methodologies.
- Use of Python libraries such as Pandas, NumPy, and Scikit-learn.
- Deployment of machine learning models using Streamlit.

KEY PROJECT / ACHIEVEMENT

- Designed and implemented an end-to-end Diabetes Prediction System.
- Applied Logistic Regression and Random Forest classifiers.
- Random Forest model demonstrated improved predictive performance.
- Developed a user-friendly interface for real-time prediction.

PROJECT OUTPUT – STREAMLIT-BASED



CHALLENGES FACED

- Presence of missing and zero values in medical data.
- Selection of the most suitable machine learning model.
- Handling class imbalance within the dataset.
- Designing a clear and user-friendly interface.

HOW CHALLENGES WERE OVERCOME

- Applied mean and median imputation during preprocessing.
- Evaluated multiple models using standard performance metrics.
- Selected Random Forest based on accuracy and reliability.
- Improved user interface through iterative testing.

CONCLUSION

- The project demonstrates the effectiveness of machine learning in healthcare.
- The developed system provides accurate and reliable diabetes prediction.
- It supports early diagnosis and decision-making.
- The system enhances understanding of applied machine learning concepts.

FUTURE SCOPE

- Integration of advanced machine learning and ensemble techniques.
- Use of deep learning models for improved accuracy.
- Deployment as a mobile or cloud-based application.
- Integration with real-time healthcare and wearable devices.

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Thank You